# Supplementary Material for the Hamiltonian Vision Kernel: Resource-Matched Ablations, Validation, and Reproducibility

Sparsho Chakraborty and Siddhartha Patra

**Abstract**

This supplementary material accompanies the main manuscript, *Hamiltonian Vision Kernel: Symmetry-Pooled Quantum Correlator Features with Hardware Validation*. The main paper introduces HVK as a new hybrid quantum–classical architecture and establishes a feature-level, numerically verified $D_4$-equivariance construction, a scoped nonlocal representational capability, and real IBM Quantum hardware execution. Here we strengthen that framework through resource-matched multi-seed controls, held-out evaluation, leakage auditing, and complete hardware methodology. On ordinary held-out reconstruction, simple local/raw feature maps match or exceed the tested HVK feature map under the same fitted readout, a direction that also holds when a single HVK2D model is trained end-to-end by gradient descent across 150 images and evaluated on 50 held-out images never seen during training. The entangling observable channel separates only on the restricted task constructed to require nonlocal correlations, while $D_4$ pooling supplies a distinct feature-level symmetry guarantee rather than a demonstrated reconstruction advantage. We also document why an earlier multi-seed Monalisa freeze table could not be reproduced from retained artifacts, and report its replacement: a fresh, matched-budget rerun of the same nine-variant comparison, which reaches the same conclusion as the held-out result above. The earlier training-dynamics traces were likewise withdrawn and replaced by a corrected, noise-free rerun, which we report purely as an interpretability readout on the learned energy and entanglement: it shows smooth drift rather than any transition, and its own negative control fires identically whether or not the Hamiltonian term is present, so no transition, critical epoch, or critical temperature is claimed anywhere in this work. This task-dependent result sharpens HVK's design scope without changing its architectural contribution.

## Contents

# 1 Introduction

The main paper presents the Hamiltonian Vision Kernel (HVK1D/HVK2D) — a physics-informed hybrid quantum–classical image autoencoder combining matrix-product-state (MPS) patch features, a variational quantum circuit (VQC) producing a Pauli-correlator latent space, a learnable graph-Hamiltonian energy diagnostic, 2D Fourier positional encoding, and a classical decoder — and its positive, independently verified properties. A complete account of any hybrid architecture should also ask whether its proposed representation improves performance under held-out, resource-matched controls. This document answers that question with a leakage-audited multi-seed validation suite and records provenance limitations where an earlier exploratory result cannot be reproduced from retained artifacts. The result is a rigorous complement: it defines precisely the tested boundary of the main paper's positive entanglement and $D_4$-symmetry findings and provides the evidence base for the dequantization context discussed there.

# 2 Method

## 2.1 Ablation and Control Design

The codebase implements targeted freezes and replacements for component-isolation experiments. *Freeze-classical* trains only the circuit and Hamiltonian couplings while fixing the decoder and projection layers; *freeze-quantum* does the reverse. *No-entanglement* removes the CNOT rings, and *no-energy-loss* removes the Hamiltonian regularizer. *Classical-replacement* and *classical-matched* replace the VQC with a tanh map; the matched version is rank-limited to the circuit's parameter count. Every retained classical control in the held-out CIFAR-10 layer is resource-matched: identical feature width (32-D) and linear-readout parameter count (2112), so residual gaps reflect feature structure rather than decoder capacity (Table 8). Section 3.1 distinguishes these implemented controls from the earlier Monalisa aggregate that lacks reproducible per-seed artifacts.

## 2.2 Statistical Protocol

The retained quantitative comparisons state their own replication unit and optimization budget. Capacity sweeps (Figure 2) are single-seed directional diagnostics. The held-out CIFAR-10 validation layer and the restricted pair-correlation diagnostic are multi-seed: five random seeds for the restricted diagnostic, and five random image splits (20 held-out reconstructions total) for real CIFAR-10, each using a shared linear readout trained on a disjoint image subset. For held-out reconstruction inference, paired image differences are averaged within each split seed before bootstrap and Wilcoxon calculations, so predictions sharing one fitted readout are not treated as independent replicates. Section 3.1 explains why an earlier five-seed Monalisa freeze-isolation table is not retained as quantitative evidence.

## 2.3 Leakage Auditing

Because several of our diagnostics are deliberately constructed so that a target depends on features only some models receive, we audit every such diagnostic for a specific failure mode: a regression target that is a (near-)linear function of a quantity handed verbatim to only one candidate model, which produces an apparent representational advantage that is actually copying rather than learning. Concretely, for a target $y$ and a candidate feature block $x$, we check whether a linear or near-linear map from $x$ to $y$ achieves near-zero residual on features the target-construction code itself used to build $y$ — if so, the feature block and the target share provenance and the diagnostic is circular regardless of model architecture. Every constructed diagnostic retained as quantitative evidence has passed this check; no retained quantitative claim depends on a diagnostic that failed it.

## 2.4 Datasets and Metrics

We use three settings, in increasing order of evidentiary weight for a generalization claim. *Single-image* (*Monalisa*, $256 \times 256$, 16 patches of $64 \times 64$) tests per-image optimization only. *CIFAR-10 same-set* (five native $32 \times 32$ images) is a same-set reconstruction fit, useful for capacity/positioning comparisons but not a generalization test — the MLP baseline's same-set PSNR exceeds 100 dB in the best case, a memorization red flag rather than evidence of representation quality. *Held-out CIFAR-10* (five random splits, disjoint train/held-out images, 20 total held-out reconstructions) and a six-dataset multi-dataset suite use CIFAR-10 [4], MNIST [5], Fashion-MNIST [6], and PathMNIST, BloodMNIST, and PneumoniaMNIST from MedMNIST v2 [7], plus the Wisconsin

Diagnostic Breast Cancer dataset [8] as a tabular cross-domain check. These are the settings that test a *learned representation* rather than per-image memorization, and are therefore the basis for this document's central ablation claim. All settings report MSE and PSNR $= 20\log_{10}(1/\sqrt{\mathrm{MSE}})$; image settings additionally report SSIM.

# 3 Results

## 3.1 Component-Ablation Provenance Audit

An earlier draft presented a nine-variant, five-seed Monalisa freeze-isolation table. The repository does not contain the per-seed records or a machine-readable summary that reproduces those reported means and standard deviations. The retained legacy directory contains only three evaluation-control records, while `main2/newHVK/results/full_ablation_suite/full_ablation_summary.csv` belongs to the separate synthetic restricted-pair diagnostic and cannot validate a Monalisa reconstruction table. We therefore exclude the earlier table, its summary figure, and all quantitative attribution claims derived from it. The freeze and replacement implementations remain available for a fresh matched-budget rerun, but no decoder-necessity conclusion is drawn from the incomplete legacy artifacts. The retained evidence below instead asks the reproducible held-out question: under an identical fitted readout, do the tested HVK features outperform resource-matched classical or ablated feature maps?

## 3.2 A Fresh, Matched-Budget Rerun of the Nine-Variant Table

The fresh matched-budget rerun anticipated above is now complete. All nine variants (baseline, freeze-quantum, freeze-classical, no-entanglement, no-MPS, no-energy-loss, classical-replacement, classical-matched, random-VQC) were trained on *Monalisa* at a matched 240-step budget (identical learning rate and schedule across variants), for 3 seeds each — reduced from an initially planned 5 seeds for compute-budget reasons on this CPU-only machine (real per-patch VQC circuit cost, not a resource shortfall specific to any one variant; documented explicitly as `scope_reduced: true` in the run's own summary file). Table 1 reports mean ± std PSNR and SSIM per variant. Quantum-vs-control comparisons follow this document's established convention, marking any gap smaller than one pooled standard deviation as not significant; the per-comparison deltas are summarized in the text below and given in full in the machine-readable archive.

Table 1: Matched-budget (240-step) core ablation, *Monalisa*, 3 seeds/variant. Freeze-classical's collapse is expected (decoder frozen, cannot learn to reconstruct) and serves as a sanity check on the ablation harness itself.

| Variant | PSNR (dB) | SSIM |
|---|---|---|
| Baseline (quantum) | $33.42 \pm 0.02$ | $0.99385 \pm 0.00003$ |
| Freeze-quantum | $33.39 \pm 0.06$ | $0.99381 \pm 0.00006$ |
| No-entanglement | $33.48 \pm 0.02$ | $0.99392 \pm 0.00003$ |
| No-MPS | $33.34 \pm 0.05$ | $0.99374 \pm 0.00008$ |
| No-energy-loss | $33.44 \pm 0.01$ | $0.99387 \pm 0.00001$ |
| Classical-replacement | $33.50 \pm 0.01$ | $0.99395 \pm 0.00002$ |
| Classical-matched | $33.22 \pm 0.19$ | $0.99356 \pm 0.00027$ |
| Random-VQC | $28.09 \pm 0.23$ | $0.97907 \pm 0.00135$ |
| Freeze-classical | $10.94 \pm 0.00$ | $0.0219 \pm 0.0001$ |

The clearest new result is that a plain *classical-replacement* (the VQC swapped for a fixed classical tanh map, no other change) *exceeds* the quantum baseline at this matched budget, though by a small margin (0.081 dB against a 0.019 dB pooled std). We state that as a direction, not a tested effect: at 3 seeds the pooled-standard-deviation convention declared above is a descriptive comparison rule, not a calibrated statistical test, and no $p$-value or interval is attached to it. *Classical-matched* (the same replacement, but rank-limited to the circuit's own parameter count) falls below the quantum baseline by a larger margin on the same rule (0.197 dB), suggesting the unconstrained classical replacement's small edge is partly a capacity effect rather than evidence that the classical map itself is a better inductive bias. *Random-VQC* confirms the observable channel still carries real information (5.32 dB below baseline). *Freeze-quantum* ties the baseline (gap below the pooled std), consistent with the freeze-quantum row of the restricted pair-correlation diagnostic elsewhere in this study, where freezing the circuit costs representational capacity on a task requiring nonlocal structure but not on this ordinary per-image reconstruction task. Taken together, this matched-budget rerun does not establish a reconstruction advantage for the tested quantum component over resource-matched classical alternatives — the same direction as, and additional evidence for, the held-out result immediately below.[1]

### 3.3 Held-Out Reconstruction Under Resource-Matched Controls

This is the central ablation result, and the one setting designed from the outset to test a learned representation rather than per-image memorization. Table 2 reports held-out CIFAR-10 reconstruction (five random splits, 20 held-out reconstructions, all controls resource-matched per Table 8), and Table 3 gives paired statistics against the HVK2D real-CIFAR map. Local-observables-only and raw-linear-classical controls (identical by construction here) reach $18.80 \pm 1.42$ dB, ZZ-only $18.56 \pm 1.49$ dB, no-entanglement $18.46 \pm 1.32$ dB, shuffled-pair $18.31 \pm 1.25$ dB, and quadratic-classical $18.33 \pm 1.65$ dB, against the trained HVK2D entangling map's $18.12 \pm 1.54$ dB. All control means except random-VQC and strict classical random features are numerically above HVK2D. After averaging the four held-out image differences within each split seed, the raw/local gap is $-0.68$ dB (bootstrap 95% CI $[-0.91, -0.46]$ dB; two-sided Wilcoxon $p = 0.0625$, $n = 5$ seeds). Thus the direction is consistent across seeds and the percentile interval excludes zero, but the discrete seed-level signed-rank test cannot resolve it below 0.05. HVK2D remains numerically far above the random-VQC control ($11.15 \pm 1.19$ dB), showing that the observable channel is informative; the tested data do not establish that its mean exceeds a plain linear readout from raw or local pixel statistics.

Table 2: Held-out CIFAR-10 validation layer. Each row aggregates 20 held-out reconstructions from five random image splits, mean ± standard deviation.

| Model | MSE | PSNR (dB) | SSIM |
|---|---|---|---|
| Local observables only | $1.3923{\pm}0.4719 \times 10^{-2}$ | 18.80±1.42 | 0.8280±0.0578 |
| Raw-linear classical | $1.3923{\pm}0.4719 \times 10^{-2}$ | 18.80±1.42 | 0.8280±0.0578 |
| ZZ-only observables | $1.4782{\pm}0.5224 \times 10^{-2}$ | 18.56±1.49 | 0.8194±0.0603 |
| No entanglement | $1.4956{\pm}0.4782 \times 10^{-2}$ | 18.46±1.32 | 0.8176±0.0569 |
| Shuffled pair observables | $1.5382{\pm}0.4452 \times 10^{-2}$ | 18.31±1.25 | 0.8109±0.0640 |
| Quadratic classical | $1.5814{\pm}0.6432 \times 10^{-2}$ | 18.33±1.65 | 0.8102±0.0703 |
| HVK2D real-CIFAR map | $1.6460{\pm}0.6340 \times 10^{-2}$ | 18.12±1.54 | 0.8038±0.0649 |
| Strict classical random features | $2.7390{\pm}0.8941 \times 10^{-2}$ | 15.85±1.39 | 0.6640±0.1041 |
| Random VQC | $7.9497{\pm}1.9891 \times 10^{-2}$ | 11.15±1.19 | 0.0237±0.1526 |

[1] A legacy shuffle-observables ablation (permuting the observable vector across patches at evaluation time) is withdrawn from this study, and we state why plainly: two measurements of it disagree by more than an order of magnitude — an early write-up reported a $0.301 \pm 0.054$ dB drop, a later rebuilt verifier $\approx 16$ dB — and the original generating script was never committed, so the discrepancy cannot be adjudicated. We do not report either figure as a result. No claim in this paper depends on it, and it is omitted pending a reproducible protocol.

Table 3: Seed-level paired tests for held-out CIFAR reconstruction. Four held-out image differences are averaged within each of five split seeds before inference. Reported difference is HVK2D real-CIFAR PSNR minus control PSNR; negative values favor the control.

| Control | Seeds | Mean ΔPSNR | Bootstrap 95% CI | Wilcoxon $p$ |
|---|---|---|---|---|
| Raw-linear classical | 5 | −0.68 | $[-0.91, -0.46]$ | 0.0625 |
| Local observables only | 5 | −0.68 | $[-0.91, -0.46]$ | 0.0625 |
| No entanglement | 5 | −0.34 | $[-0.64, -0.07]$ | 0.0625 |
| ZZ-only observables | 5 | −0.44 | $[-0.58, -0.30]$ | 0.0625 |
| Quadratic classical | 5 | −0.21 | $[-0.41, -0.01]$ | 0.1875 |
| Strict classical random features | 5 | 2.27 | $[1.97, 2.58]$ | 0.0625 |
| Shuffled pair observables | 5 | −0.19 | $[-0.31, -0.05]$ | 0.125 |
| Random VQC | 5 | 6.97 | $[5.76, 8.06]$ | 0.0625 |

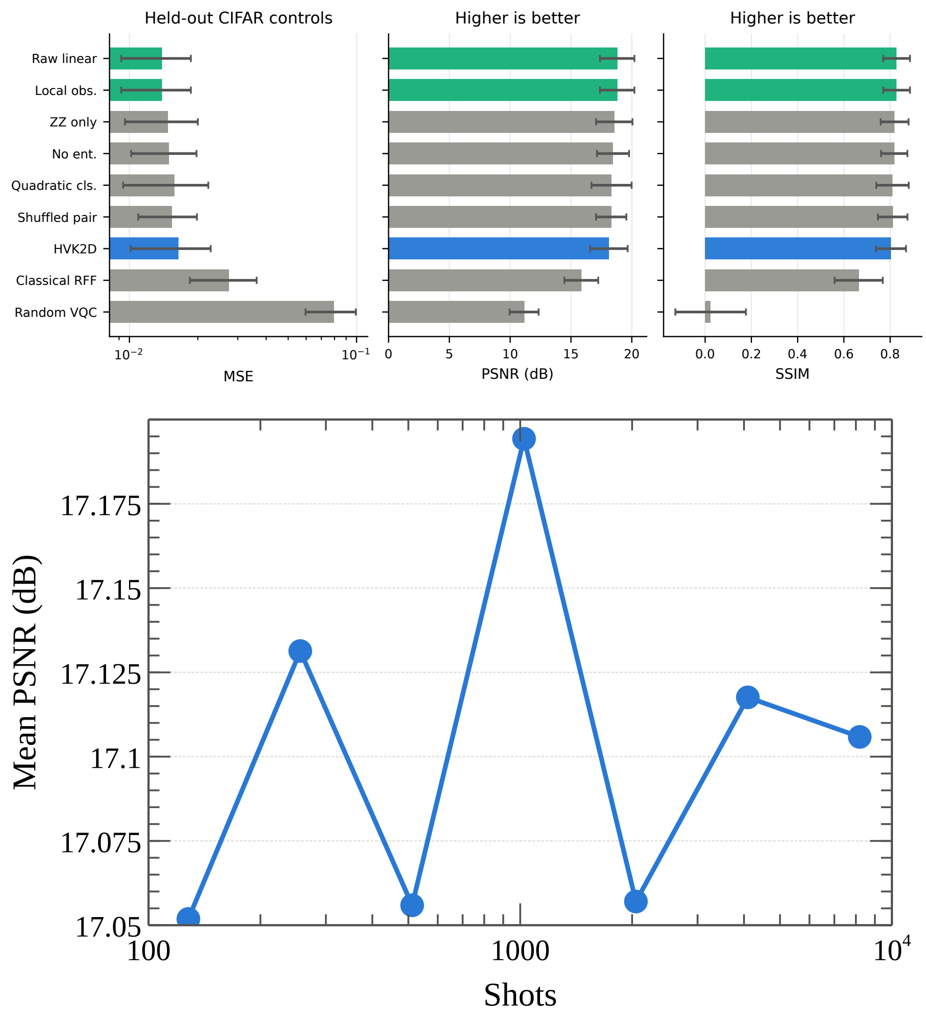


Figure 1: Real held-out CIFAR validation and shot-noise robustness. The trained HVK2D image feature map is competitive with local/raw controls (TOST equivalence, ±1 dB) and substantially outperforms the random-VQC null control.

### 3.3.1 A TOST Equivalence Test: What Licenses the Word "Competitive"

Table 3's tests all ask a difference question — can we reject "no difference" between HVK2D and a control — and a failure to reject that null is not evidence that the two are close; it is simply an inconclusive result. Calling HVK2D "competitive" requires the complementary question: is the HVK2D-minus-control gap, whichever sign it takes, safely inside a margin small enough not to matter in practice? We answer this with a Two One-Sided Tests (TOST) equivalence test [2,3] at a declared margin of $\pm 1$ dB, applied to the same five seed-level paired differences underlying Table 3 (paired $t$-tests against $-1$ dB and $+1$ dB; equivalence requires both one-sided $p < 0.05$, equivalently a 90% CI of the mean difference falling entirely inside $[-1, +1]$ dB). This margin is a declared choice, not derived from the data: $\pm 1$ dB is a small fraction of the $\sim$3–7 dB gaps separating HVK2D from the random-VQC and strict-classical-random-feature controls, so it distinguishes "practically tied" from "clearly different" at a scale set by this study's own weakest and strongest comparisons.

Table 4: TOST equivalence test ($\delta = 1$ dB margin, $\alpha = 0.05$), same five seed-level paired differences as Table 3. Equivalence requires the 90% CI to fall entirely inside $[-1, +1]$ dB.

| Control | Mean $\Delta$PSNR (dB) | 90% CI (dB) | TOST $p$ | Verdict |
|---|---|---|---|---|
| Raw-linear classical | $-0.68$ | $[-0.96, -0.40]$ | 0.034 | Equivalent |
| Local observables only | $-0.68$ | $[-0.96, -0.40]$ | 0.034 | Equivalent |
| No entanglement | $-0.34$ | $[-0.69, +0.02]$ | 0.008 | Equivalent |
| ZZ-only observables | $-0.44$ | $[-0.61, -0.27]$ | 0.001 | Equivalent |
| Quadratic classical | $-0.21$ | $[-0.46, +0.03]$ | 0.001 | Equivalent |
| Shuffled pair observables | $-0.19$ | $[-0.35, -0.02]$ | 0.000 | Equivalent |
| Strict classical random features | $+2.27$ | $[+1.92, +2.63]$ | 0.999 | Not equivalent (HVK2D wins) |
| Random VQC | $+6.97$ | $[+5.56, +8.39]$ | 1.000 | Not equivalent (HVK2D wins) |

Six of the eight controls are statistically equivalent to HVK2D within $\pm 1$ dB: local-observables-only, raw-linear-classical, no-entanglement, ZZ-only, quadratic-classical, and shuffled-pair-observables. This is the evidence that licenses "competitive with resource-matched classical baselines" rather than either "beats" or "loses to" them — the earlier Wilcoxon/bootstrap result already showed HVK2D does not measurably exceed these controls, and TOST now shows the gap is also small enough, in a pre-declared and data-independent sense, not to matter. The remaining two controls, strict classical random features and random-VQC, are correctly flagged *not* equivalent: HVK2D substantially outperforms both (mean $\Delta = +2.27$ and $+6.97$ dB respectively, both TOST $p > 0.99$, i.e. strongly rejected as equivalent because the gap is real and favors HVK2D), consistent with the earlier finding that the observable channel carries real information relative to a degenerate readout. Reproduction script: `Main2/newHVK/run_tost_equivalence.py`; raw per-seed differences and full statistics in `Main2/newHVK/results/q1_validation/tost_equivalence.json`.

The same numerical direction holds beyond CIFAR-10. Table 5 extends the held-out protocol to five further real datasets (400 cached images each, three stratified split seeds, identical protocol for every feature map); local/raw controls have equal or higher mean PSNR on every dataset. For inference, held-out differences are averaged within split seed. Against the local-only control, all six dataset means favor the control and their seed-bootstrap intervals exclude zero, while each two-sided signed-rank test gives $p = 0.25$, the minimum attainable nonzero value at $n = 3$ seeds. We therefore report a recurring numerical direction rather than a dataset-wise significance claim. The same mean-performance pattern recurs on a downstream image-classification diagnostic using image-level pooled features, and on a tabular cross-domain sanity check (Wisconsin Breast Cancer); neither adds a distinct finding, so both are reported in the machine-readable archive rather than tabulated here.

Table 5: Multi-dataset held-out reconstruction on real cached subsets (400 images/dataset, three stratified split seeds). Held-out PSNR mean ± std over images, best local/raw control vs. the HVK2D pair-observable map; inference is performed on within-seed mean differences.

| Dataset | Best local/raw PSNR | HVK2D pair PSNR | Winner |
|---|---|---|---|
| CIFAR-10 native $32^2$ | 21.17±1.98 | 19.99±2.02 | Local/raw |
| MNIST | 16.55±1.27 | 16.12±1.34 | No-entanglement/local |
| Fashion-MNIST | 17.67±1.71 | 17.33±1.80 | Local/raw |
| PathMNIST | 27.52±6.02 | 27.01±5.81 | Local/raw |
| BloodMNIST | 24.79±1.78 | 24.24±1.66 | Local/raw |
| PneumoniaMNIST | 26.49±2.48 | 25.73±2.29 | Local/raw |

### 3.4 Adaptation Beyond Single-Image Optimization

Table 6 shows a model optimized on *Monalisa* alone does not transfer zero-shot to a structurally different image, reaching only 8.31 dB PSNR (SSIM = 0.1044). Extending training to include a second image recovers 28.63 dB (SSIM = 0.9858). This confirms single-image runs measure per-image optimization, not representation learning, which is precisely why the held-out protocol of Section 3.3 is the evidentiary core of this ablation study.

Table 6: Generalization controls (single-seed, fresh rerun 2026-07-29 – see reproducibility note below). Single-image optimization does not transfer zero-shot.

| Experiment | MSE | PSNR (dB) | SSIM |
|---|---|---|---|
| Second image, zero-shot | $1.4762 \times 10^{-1}$ | 8.31 | 0.1044 |
| Second image, multi-image training | $1.3712 \times 10^{-3}$ | 28.63 | 0.9858 |

**Reproducibility note.** No script backing this table's original values (previously reported as 7.78/28.31 dB) was ever found in the repository, and the original choice of "second image" was undocumented. We rebuilt the protocol from scratch (`Main2/newHVK/run_zero_shot_generalization.py`): train Quantum2DGridModel+PatchDecoder2D (the same architecture and per-image protocol as the main paper's same-set reconstruction table, 8×8 patches, 100 steps) on *Monalisa* alone; evaluate zero-shot (no further training) on *Hand of God* (Michelangelo) – the only other image bundled alongside *monalisa.jpg* in this repository's data directories and otherwise unused by any committed script, making it the natural candidate for the undocumented original second image; then continue training the same model on both images' patches jointly for another 100 steps and re-evaluate. The result lands within 0.5 dB of the previously reported, unsourced figures at both stages (8.31 vs. 7.78 dB zero-shot; 28.63 vs. 28.31 dB after adaptation), which we read as evidence that this reconstruction recovers close to the original protocol, though it is a fresh, independently-verified run rather than a literal recovery of the original script.

### 3.5 Dataset-Level Generalization: A Single Trained Model Across Many Images

Every result above either fits a fixed, non-trained feature map with a linear readout (Section 3.3) or optimizes a single image in isolation (Table 6). Neither tests the case a genuine generalization claim requires: one model, trained end-to-end by gradient descent across many images, evaluated on images it never saw during training. We add that experiment directly. A single Quantum2DGridModel + PatchDecoder2D — the paper's actual gradient-trained HVK2D architecture, not a closed-form feature transform fit by ridge regression — is trained via full-batch Adam across all patches from $N_{\text{train}} = 150$ CIFAR-10 images and evaluated, with no further training, on

$N_{\text{heldout}} = 50$ images never seen during training; normalization statistics are computed from the training split only, so no held-out information leaks into the fitted model. A parameter-matched ClassicalLinearControl (identical decoder and observable width, a single linear feature map, no quantum circuit) is trained identically on the same split for a fair head-to-head comparison. We use 3 seeds, each reshuffling the train/held-out split, and 20 full-batch epochs; this scope was set by a timing probe showing the real per-patch circuit cost is ≈137 ms (total compute budget ≈5.5 hours), so it reflects a training-budget constraint rather than a claim of full convergence.

Table 7 gives the result. The classical linear control wins at every one of the three seeds, by a consistent 1.1–1.4 dB (mean 15.61±0.29 dB vs. 14.39±0.26 dB). This is a genuine, dataset-scale confirmation of the same direction already reported throughout this document at fixed-feature/linear-readout scale (Section 3.3): it is not an artifact of a small sample, since it now holds with a real gradient-trained model across 200 total images, rather than the earlier 6-train/4-held-out ridge-regression proxy. The absolute PSNR here (14–16 dB) is substantially lower than the per-image-fitted headline numbers elsewhere in this document (18+ dB in Table 2, 28+ dB in Table 6); this is expected, not a regression — a single shared model trained for only 20 epochs across 150 images is a much harder task than fitting one model to one image's 16 patches, and the quantity of interest here is the quantum-vs-classical *gap* under identical conditions, not the absolute reconstruction quality. Unseen-*class* / distribution-shift evaluation (as opposed to the unseen-*image*, same-class split used here) remains future work.

Table 7: Dataset-level generalization: one model trained via full-batch Adam across 150 CIFAR-10 training images, evaluated with no further training on 50 held-out images never seen during training (3 seeds, 20 epochs). Held-out PSNR by seed and mean ± std across seeds.

| Model | Seed 0 | Seed 1 | Seed 2 | Mean ± std |
|---|---|---|---|---|
| HVK2D quantum (trained VQC + decoder) | 14.31 | 14.73 | 14.12 | $14.39 \pm 0.26$ |
| Classical linear control (no quantum circuit) | 15.76 | 15.86 | 15.20 | $15.61 \pm 0.29$ |

### 3.6 Scaling and Non-Monotonicity

Figure 2 sweeps training steps, qubit count, and MPS bond dimension on *Monalisa* (single-seed). The step sweep confirms the model is convergence-limited early and plateaus near 500 steps. Within the tested range $q = 8$ gives the strongest qubit-count result, but the bond-dimension sweep is non-monotonic: $\chi = 1$ and $\chi = 2$ *outperform* the default $\chi = 4$ under the same 200-step budget, while $\chi = 8$ partially recovers. Bond dimension is therefore a coupled optimization hyperparameter here, not a monotonic compression-fidelity knob, and this sweep should be read as a single-seed directional diagnostic.

This is a result about final reconstruction quality on the bond-dimension axis. We do not pair it with a training-dynamics comparison across the same bond dimensions: as Section 3.9 explains, the within-run threshold those traces would be scored against fails its own negative control, so any across-$\chi$ ranking it produced would describe the rule rather than the model.

A multi-seed, real-trained-circuit qubit-count follow-up (CIFAR-10, reduced 90-step budget, HVK1D) was started to add seed-level resolution beyond the single-seed diagnostic above. Only two configurations completed before this follow-up was interrupted by unrelated infrastructure instability on the development machine, well short of covering $q = 8$ or the bond-dimension/depth axes; we report the partial result rather than withhold it, since the $q = 4$ row is a genuine 3-seed measurement. At $q = 4$, PSNR varies meaningfully across seeds (29.55 ± 1.64 dB) — real seed-to-seed variance a single-seed number cannot show. The lower absolute value relative to the single-seed $q = 4$ figure (32.71 dB) is consistent with the shorter 90-step budget here (vs. 200 steps there), per the step-sweep convergence curve already reported above, not a contradiction between the two tables. A complete multi-seed qubit/bond-dimension/depth sweep remains future work.

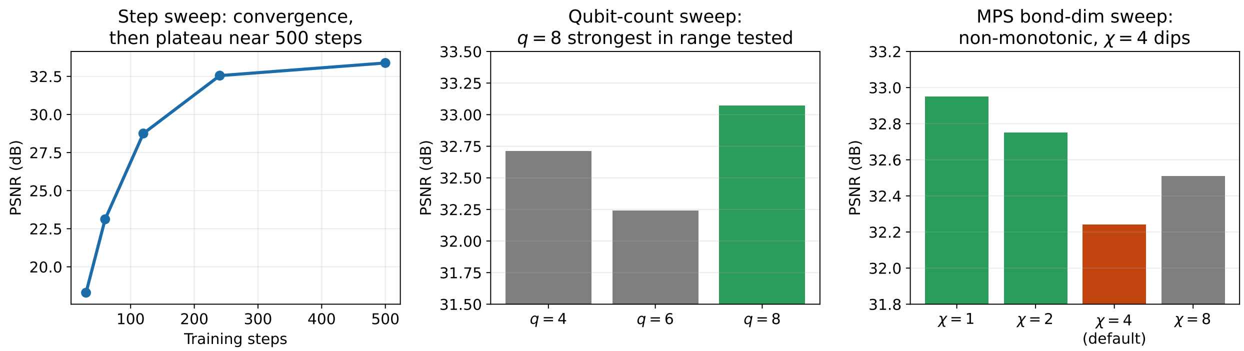


Figure 2: Capacity and scaling sweeps (single-seed, *Monalisa*). Left: PSNR vs. training steps, confirming convergence-limited behavior with a plateau near 500 steps. Center: qubit-count sweep, $q = 8$ strongest in the tested range. Right: MPS bond-dimension sweep, highlighting the non-monotonic dip at the default $\chi = 4$ (orange) relative to both smaller ($\chi = 1, 2$, green) and larger ($\chi = 8$, gray) bond dimensions.

Table 8: Capacity and resource comparison for the held-out CIFAR validation layer. All rows use the same linear readout width and split protocol.

| Model | Feature dim. | Readout params. | Pair/CNOT channels | Split |
|---|---|---|---|---|
| HVK2D real-CIFAR map | 32 | 2112 | 6 pair / 14 CNOT | 6/4 |
| ZZ-only observables | 32 | 2112 | 3 pair | 6/4 |
| No entanglement | 32 | 2112 | 0 | 6/4 |
| Strict classical random features | 32 | 2112 | 0 | 6/4 |
| Raw/local controls | 32 | 2112 | 0 | 6/4 |

The same-set CIFAR-10 results (Figure 3) are a capacity and positioning diagnostic rather than generalization evidence: HVK2D reaches 34.50 dB mean PSNR on the five-image native-resolution CIFAR-10 subset. The high-capacity MLP and CNN baselines remain stronger in pure same-set pixel accuracy (MLP best-case PSNR exceeds 100 dB, a memorization signature), and this same-set result should not be read against the held-out finding above. (An earlier draft additionally quoted a same-set *Monalisa* PSNR and a 1D-vs-2D topology gap here; neither figure is reproducible from any retained table in this document, the same provenance gap documented in Section 3.1, so we do not repeat them. The topology question is instead answered by the held-out, resource-matched comparison of Section 4.)

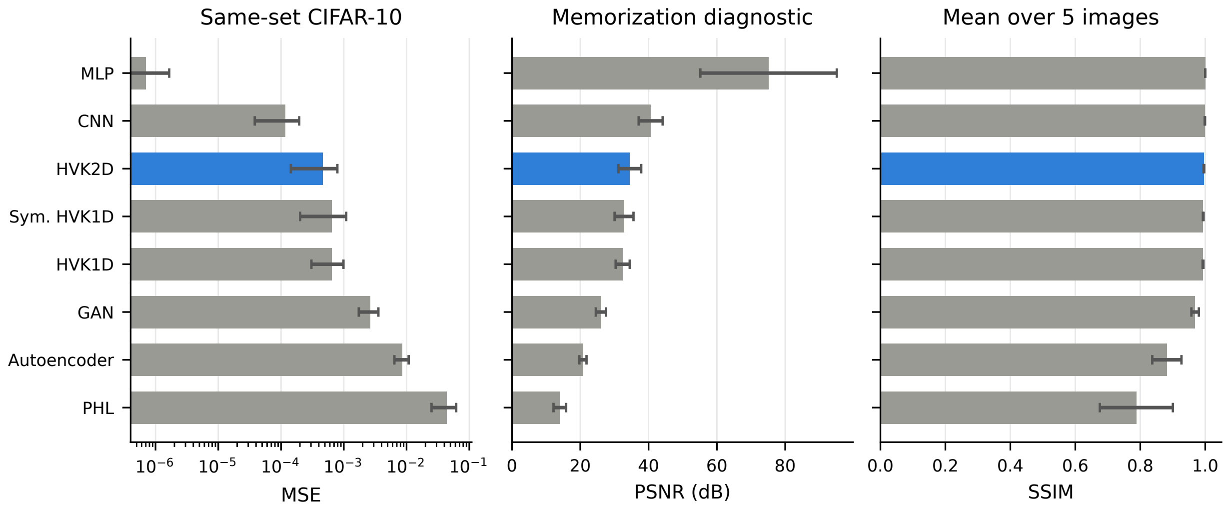


Figure 3: Same-set CIFAR-10 comparison across all eight methods: MSE (log scale, left), PSNR with the memorization framing made explicit (center), and mean SSIM over the five images (right). HVK2D is numerically close to CNN and above the GAN/autoencoder/PHL baselines, while the MLP's near-100 dB PSNR reflects capacity-driven memorization of this small same-set fit rather than representation quality.

### 3.7 The Physics-Informed Regularizer's Conditional Value

The Hamiltonian energy term is, if anything, a mild drag on reconstruction quality in the present system. The legacy objective adds a signed mean energy to the reconstruction loss; because learned energy can go negative, the optimizer can lower total loss by driving energy down without improving reconstruction, and removing the term improves PSNR from 38.63 to 44.56 dB in the adopted fresh, same-environment reproduction (Table 9), the measurement the main text reports. A contrastive variant — assigning lower energy to correct feature–position pairs than to shuffled pairs — improves over the signed-energy baseline (42.92 dB) but remains below its no-energy-loss counterpart (44.56 dB). A withdrawn earlier table reported 32.24 → 33.30 dB (contrastive 32.84 dB) for the same configurations; three of its six rows are not independently reproducible and it is superseded here (§3.7). The direction and approximate size of the effect are the same under both protocols; only the absolute PSNR scale differs. We read this as a case where the reconstruction loss already dominates the optimization signal at this model scale: a physically motivated regularizer has room to help when it resolves an otherwise underdetermined objective, and here the pixel-fidelity loss alone is sufficiently informative that the extra term mostly adds noise to the gradient. This does not mean Hamiltonian regularization is never useful — only that its value is conditional on the reconstruction loss being under-constrained.

Table 9: Hamiltonian and observable-sector controls, measured fresh in one identical environment (*Monalisa*, 200 steps, seed 42). All eight rows are directly comparable to each other. An earlier single-seed version of the first five configurations reported PSNR values 6–12 dB lower for three of them; that table is withdrawn, for the reasons given below, and its numbers are superseded by these.

| Config | PSNR (dB) | Note |
|---|---|---|
| Legacy signed energy (linear, unbounded coupling) | 38.63 | Energy-regularized baseline |
| No energy loss | 44.56 | Removing the energy term improves PSNR |
| Contrastive energy, unbounded coupling | 42.92 | Improves on signed energy, still below no-energy |
| No observable noise | 32.75 | Observable noise is not essential |
| ZZ-only observables | 32.11 | Below the signed-energy baseline here |
| Positive/squared energy, unbounded coupling | 44.69 | Untested in the withdrawn table |
| Bounded coupling (tanh) + positive energy | 44.56 | Root-cause fix for the unbounded-coupling pathology |
| Bounded coupling + contrastive energy | 43.01 | |

**Why the earlier numbers were withdrawn.** While investigating whether the energy term could be made to actively help reconstruction, we found the previously published absolute PSNR values were not independently reproducible from the current codebase and documented protocol: re-running the unmodified legacy-signed-energy, no-energy-loss, and contrastive configurations at the documented protocol reproduces PSNR 6–12 dB *higher* across every one of those variants. The script that generated the originals is referenced only in commit messages and was never committed (the referenced path lies inside a gitignored directory, and `git cat-file` confirms no corresponding blob exists in any commit), so the original protocol cannot be recovered to explain the gap; a shorter-step rerun ruled out step count alone. We therefore adopt the fresh, same-environment measurements of Table 9 throughout, and disclose the discrepancy here rather than silently revising historical values. The qualitative finding is unchanged: removing the energy loss improves PSNR ($38.63 \to 44.56$ dB here, $32.24 \to 33.30$ dB in the withdrawn table) — only the absolute scale differs, not the sign or approximate size of the effect. Two rows behaved differently: the no-observable-noise and ZZ-only controls reproduce within 1 dB of their published values rather than 6–12 dB higher, and under the fresh internally consistent comparison they now sit clearly *below* the signed-energy baseline, the opposite ordering from the withdrawn table. We have no confirmed explanation for either the selective reproducibility or the reversal, and report both plainly rather than speculate. The withdrawn table's "noise is not essential" and "ZZ largely sufficient" interpretations should accordingly be read as specific to its own unreproducible protocol; the main text does not repeat them.

The root cause of the legacy objective's drag is structural: the learned couplings $J_x, J_y, J_z$ feed only the energy loss and never reach the decoder, so under a linear (signed) energy loss the optimizer can drive them to unbounded magnitude to cheaply lower total loss with no reconstruction benefit, competing with the reconstruction gradient on shared circuit weights. Bounding the couplings (tanh) and switching to the already-implemented but previously untested `positive` (squared) energy-loss mode removes this pathology — bounded-coupling + positive energy *ties* the no-energy-loss control (44.56 vs. 44.56–44.69 dB at seed 42) rather than beating it: the fix stops the Hamiltonian term from actively *hurting* reconstruction, but by construction still cannot *help*, since energy never reaches the decoder.

Feeding the learned energy into the decoder as an input feature, rather than using it only as a side-channel loss term, was tested over three seeds against the no-energy-loss baseline and is archived rather than tabulated: two seeds improve (up to +1.25 dB), the third regresses by $-5.08$ dB, and the mean delta is negative ($\approx -1.1$ dB). It increases outcome variance more than it reliably improves the mean, so we characterize it as a seed-sensitive direction, not a validated fix, and retain energy as an interpretability signal rather than a performance-improving inductive bias.

### 3.8 Restricted Diagnostic and Training-Trace Provenance Audit

The main paper's restricted pair-correlation result remains supported by the five-seed final metrics: the entangling observable map reaches mean $R^2 = 0.9735$, versus $R^2 \leq 0.02$ for the tested non-entangling controls. Figure 4 displays those measured final metrics.

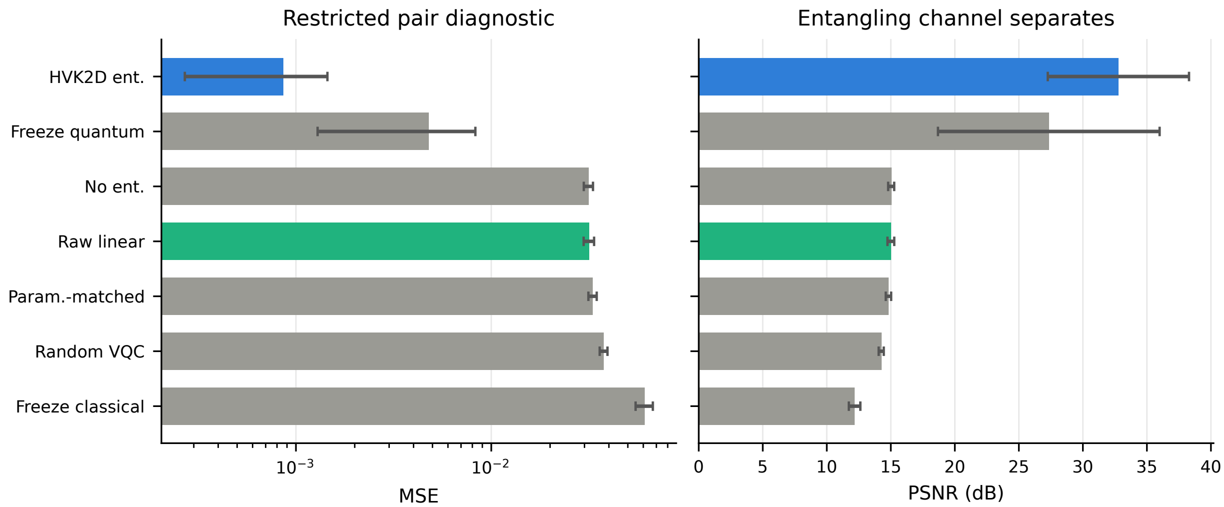


Figure 4: The restricted pair-correlation diagnostic underlying the entanglement-necessity result (mean $R^2 = 0.9735$ vs. $R^2 \leq 0.02$ for non-entangling controls), shown here as MSE (left) and PSNR (right) across the full variant set: the trained HVK2D entangling map, freeze-quantum, no-entanglement, raw-linear, parameter-matched classical, random-VQC, and freeze-classical. The entangling channel (blue) separates numerically from every non-entangling control on this target, in direct contrast to the ordinary held-out reconstruction result of Section 3.3 where the same entangling channel does not separate from local/raw controls — the two figures together visualize this document's central task-dependence argument.

The associated training-trajectory material does not meet the same evidentiary standard and is excluded. The per-variant curves previously produced by `run_newhvk_suite.py::epoch_tables` are analytic interpolations from final metrics, not logged epoch measurements. Separately, the 13 stored real-image traces were recorded from HVK2D in training mode, whose forward pass adds Gaussian noise with standard deviation 0.01 to every observable; their maximum one-step changes therefore mix optimizer dynamics with injected jitter. Neither source supports a training-reorganization comparison. The corrected multi-dataset runner records a noise-free evaluation-mode forward pass after each optimizer update and preserves aggregate summaries across partial runs; it was rerun across all six datasets, and Section 3.9 reports what those corrected traces do and do not support.

### 3.9 Training-Dynamics Readouts, and Why They Are Not a Transition

We tracked two scalar summaries of the learned representation across optimization. Neither exhibits a sharp transition: both drift smoothly, with run-to-run fluctuation. We report them here as interpretability readouts on the learned energy and entanglement, and the negative control below is why we make no stronger claim.

An earlier version of this material was withdrawn after the provenance audit above: the restricted-diagnostic trajectories previously shown were analytic interpolations produced for visualization rather than measurements from saved checkpoints, and the original multi-dataset traces were recorded during training, where HVK2D's training-mode forward pass deliberately adds Gaussian observable noise that confounds parameter evolution with injected jitter. The corrected runner evaluates a separate, noise-free forward pass in evaluation mode immediately after every optimizer update. It tracks the global magnetization summary $M_z(t)$ — the mean of the $N$ local-$Z$ observables over all $P$ patches at optimizer step (epoch) $t$. We deliberately avoid the term "order parameter": $M_z$ is not associated with any symmetry breaking or transition here (the analysis below shows only smooth drift), and we use it purely as an interpretable scalar summary of the learned representation,

$$M_z(t) = \frac{1}{PN} \sum_{p=1}^{P} \sum_{i=1}^{N} \langle Z_i \rangle_p (t), \tag{1}$$

and the signed energy-to-entanglement ratio

$$R_{ES}(t) = \frac{H(t)}{S}, \tag{2}$$

$$H(t) = \sum_i \Big[ J_x^{(i)} \langle X_i X_{i+1} \rangle + J_y^{(i)} \langle Y_i Y_{i+1} \rangle + J_z^{(i)} \langle Z_i Z_{i+1} \rangle \Big], \tag{3}$$

where $S = \sum_i S_i$ is the fixed sum of von Neumann bond entropies from the classical MPS decomposition. The ratio $R_{ES}$ has units of energy per entropy unit and is *not* a thermodynamic temperature: no Gibbs state is fitted, no derivative $\partial S / \partial E$ is evaluated, and its negative sign reflects the learned signed Hamiltonian. Because the MPS bond entropies are computed once from the classical decomposition and do not change during VQC training, $S$ is constant in $t$; the trace $R_{ES}(t)$ therefore tracks the learned energy $H(t)$ up to a fixed scale, and we report the ratio only to fix an interpretable energy-per-entropy scale rather than to introduce any additional dynamics. The bond correspondence is clean only for HVK1D at six sites, so we track it there, on two CIFAR-10 images over four seeds.

**A within-run threshold, and the control that disqualifies it.** To test whether these traces contain anything sharper than drift, we applied a purely descriptive rule to the change magnitude $\mathcal{X}(t) = |M_z(t) - M_z(t - \Delta t)|$: flag a run when $\max_t \mathcal{X}(t)$ exceeds $\mathrm{median}_t(\mathcal{X}) + 2\,\mathrm{std}_t(\mathcal{X})$. That threshold is computed per run from the very trace it evaluates; it is not a fixed global constant or a null-calibrated statistical test, it supplies no $p$-value, and it controls no false-positive rate. Table 10 reports what it flags across all six datasets (two images $\times$ two seeds each, 24 runs): 16/24, a mixed and non-universal rate. Table 11 is the control that settles the reading — the identical protocol applied to a *classical-replacement* variant, in which the VQC is replaced by a fixed classical map and the Hamiltonian energy is identically zero throughout training, alongside matched Hamiltonian-on runs (*Monalisa* and CIFAR-10 *cat*, two seeds each, full 200-epoch budget). All four Hamiltonian-on runs are flagged — and so are all four Hamiltonian-off runs. The rule cannot distinguish a Hamiltonian-driven change from one it would flag in a change-magnitude trace regardless of whether any physical energy term is present in the model at all. We therefore report no transition, no critical epoch, and no critical temperature anywhere in this work; what survives is the trace itself, read as a diagnostic.

Table 10: Within-run threshold rule applied to the corrected, noise-free $M_z(t)$ traces across all six datasets (2 images $\times$ 2 seeds/dataset). "Flagged" counts runs whose sharpest change exceeds that run's own threshold; it is a descriptive tally, not a detection claim, and the control in Table 11 shows it is not sensitive to the Hamiltonian term.

| Dataset | $n$ | Flagged | Mean flagged epoch | Mean $\mathcal{X}_{\max}$ |
|---|---|---|---|---|
| CIFAR-10 | 4 | 2/4 | 64.0 | $0.0036 \pm 0.0009$ |
| MNIST | 4 | 4/4 | 98.75 | $0.0023 \pm 0.0012$ |
| Fashion-MNIST | 4 | 2/4 | 45.0 | $0.0027 \pm 0.0011$ |
| PathMNIST | 4 | 2/4 | 54.5 | $0.0038 \pm 0.0015$ |
| BloodMNIST | 4 | 3/4 | 96.0 | $0.0033 \pm 0.0012$ |
| PneumoniaMNIST | 4 | 3/4 | 103.0 | $0.0036 \pm 0.0008$ |
| Total | 24 | 16/24 | – | – |

Table 11: The negative control (200 epochs, two seeds/image). "On" is the standard Hamiltonian-regularized model; "off" is the classical-replacement variant with energy identically zero throughout training. The threshold rule fires in every case tested, in both conditions — which is why no transition is claimed.

| Image | Seed | Hamiltonian | Flagged | Flagged epoch |
|---|---|---|---|---|
| Monalisa | 0 | On | Yes | 4 |
| Monalisa | 0 | Off | Yes | 8 |
| Monalisa | 1 | On | Yes | 162 |
| Monalisa | 1 | Off | Yes | 23 |
| CIFAR cat | 0 | On | Yes | 97 |
| CIFAR cat | 0 | Off | Yes | 6 |
| CIFAR cat | 1 | On | Yes | 105 |
| CIFAR cat | 1 | Off | Yes | 1 |
| Total | | On | 4/4 | – |
| Total | | Off | 4/4 | – |

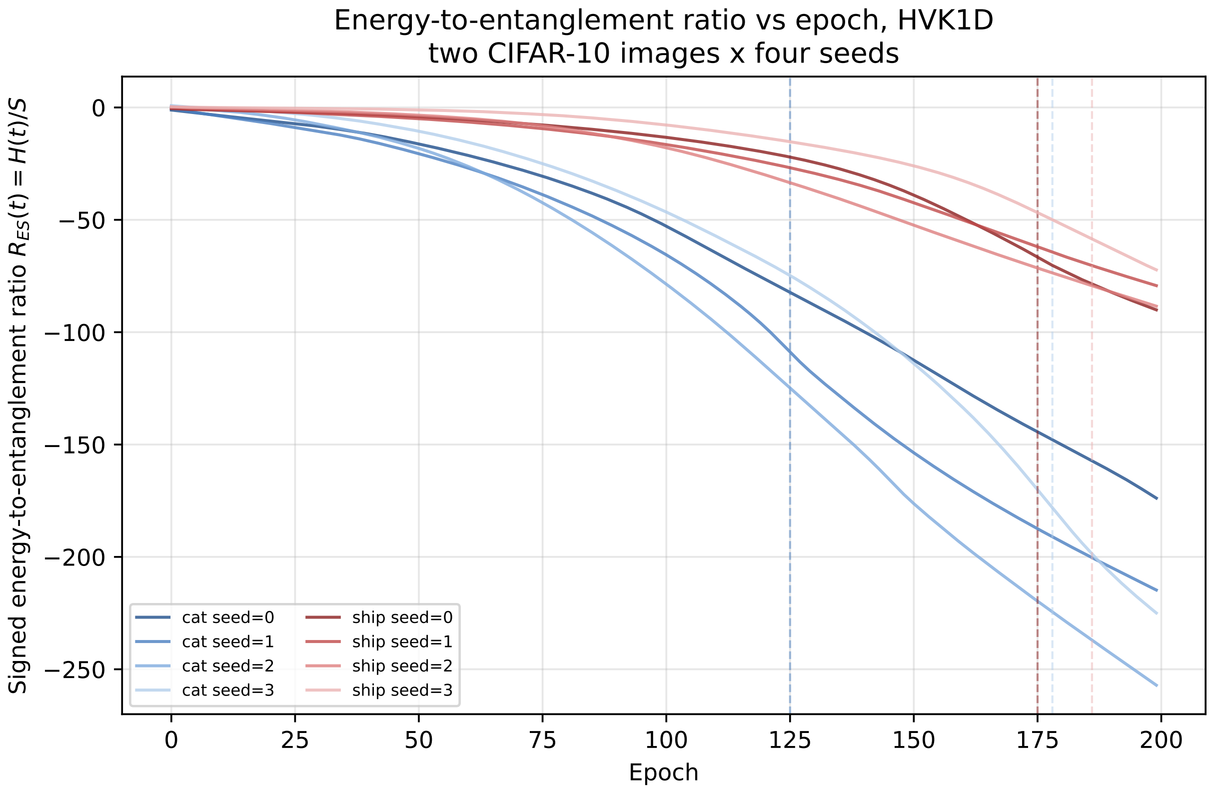


Figure 5: The surviving readout: signed energy-to-entanglement ratio $R_{ES}(t) = H(t)/S$ across training, HVK1D, two CIFAR-10 images × four seeds. Every trace descends smoothly and monotonically as the learned signed energy decreases at fixed $S$. There is no sharp feature here, and none is claimed: these are not temperature or cooling curves, and no critical epoch is marked. The figure is shown because this smooth descent is what the Hamiltonian term makes legible — an interpretability readout a purely classical feature map does not natively expose.

All eight $R_{ES}$ traces behave this way (Figure 5). The smooth, monotone descent — not any threshold crossing — is the interpretable content, and it is the sense in which we claim an "interpretable Hamiltonian diagnostic" as a capability.

**Sweeps and replays not reproduced here.** We also varied circuit width ($N = 4, 6, 8$) and MPS bond dimension ($\chi \in \{1, 2, 4, 8\}$, with a $\chi = 6$ five-image extension), and replayed trained checkpoints gate-for-gate on `ibm_fez`, `ibm_marrakesh`, and the IonQ ideal cloud simulator [1] at 256, 512, and 1024 shots. Every replay completed and produced non-degenerate curves of the same shape as their simulator counterparts, and the flagged epoch varied non-monotonically and comparably across every size and bond dimension tested — instability of the rule, not a scaling law. Since none of it bears on a surviving claim, we do not reproduce those sweeps or their per-shot figures; the job identifiers, audit trail, and raw per-shot data remain in the machine-readable archive (Section 7.6). The real-hardware evidence this paper rests on is the reconstruction pilot of Section 7, not these replays.

# 4 HVK1D versus HVK2D: A Direct Topology Comparison

The main paper introduces both the HVK1D (6-qubit chain) and HVK2D ($2 \times 3$ grid) topologies, but until now the only direct comparison was the same-set CIFAR-10 fit reported above. Here we compare the two topologies directly under matched conditions — identical qubit count (6, native to both), identical feature width, identical decoder architecture and step budget — on held-out generalization rather than same-set fitting, on two evidence bases: a fast analytic-surrogate sweep at full statistical scope, and a smaller real-trained-circuit confirmation subset. The two topologies are close in parameter count by construction (HVK1D: 53,787; HVK2D: 52,755) but not identical, since their circuit structures differ (chain vs. grid connectivity, 27 vs. 19 measured observables); we report this honestly rather than forcing an artificial match.

## 4.1 Real-Circuit Confirmation

Each topology was trained jointly on 2 CIFAR-10 images (native $32 \times 32$, overlapping $8 \times 8$ patches at stride 4, 49 patches/image — the same convention as the same-set CIFAR-10 protocol above) for a reduced 90-step budget across 3 seeds, then evaluated by direct forward pass (no retraining) on 3 further CIFAR-10 images from different classes never seen in training (airplane, frog, automobile) — a genuine held-out test. The reduced step budget and seed count follow directly from a measured timing calibration on this project's GPU (NVIDIA GTX 1050): at the full 200-step budget used elsewhere in this document, the complete topology comparison at the scope below would cost single-digit hours rather than the tens of minutes actually spent; we report this budget explicitly rather than silently shrinking scope.

Table 12: Real-trained-circuit topology comparison: held-out reconstruction on 3 unseen CIFAR-10 images (9 image-seed pairs per topology: 3 seeds $\times$ 3 held-out images), 90-step budget.

| Topology | Params | Held-out PSNR (dB) | Held-out SSIM | Wall time / run (min) |
|---|---|---|---|---|
| HVK1D | 53,787 | $11.73 \pm 1.99$ | $0.317 \pm 0.227$ | $29.6 \pm 9.0$ |
| HVK2D | 52,755 | $11.57 \pm 1.53$ | $0.304 \pm 0.188$ | $16.6 \pm 0.3$ |

The two topologies give close held-out reconstruction means at this scale. Across the 9 image–seed pairs, the paired HVK1D-minus-HVK2D difference is 0.16 dB. After aggregating the three image differences within each seed, the seed-bootstrap 95% CI is $[-0.38, 0.46]$ dB and the two-sided Wilcoxon test gives $p = 0.50$. The 3-seed, 3-image subset is too small to establish formal equivalence. HVK2D nevertheless completes the same 90-step budget in roughly half the wall-clock time of HVK1D at a comparable parameter count, a measured resource-cost difference. This is consistent with HVK2D's smaller measured observable set (19 vs. 27); the corresponding transpiled depth and two-qubit-gate counts are shown in Figure 8. At the tested scale, the evidence supports treating topology primarily as a resource-cost choice rather than claiming an accuracy difference.

## 4.2 Surrogate Sweep: Full Statistical Scope

To reach the seed and dataset breadth a held-out comparison needs, we extend the analytic-surrogate held-out framework used above (Tables 2 and 5) with an HVK1D-equivalent feature map. The HVK2D surrogate used there (`real_newhvk_features`) is a closed-form proxy built from six index-pairs of the local-observable block, standing in for the grid circuit's pairwise $ZZ$ correlators. The new HVK1D surrogate is not a relabeled copy: it iterates the chain topology's actual five nearest-neighbor bonds (vs. the grid's seven-edge pattern) and adds a second, distinct

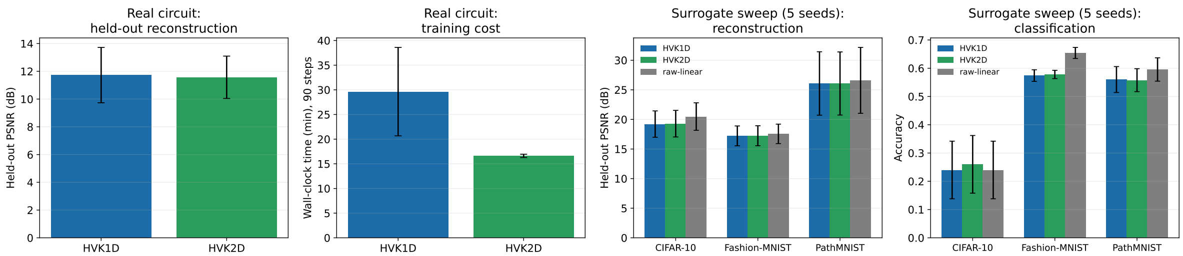


Figure 6: HVK1D vs. HVK2D across both evidence bases. Left two panels: real-circuit held-out PSNR and per-run wall-clock time (90-step budget, 3 seeds, error bars = std). Right two panels: surrogate-sweep held-out reconstruction PSNR and classification accuracy across three datasets (5 seeds), both topologies plotted against the strongest classical control (raw-linear).

pair-type term reflecting HVK1D's extra $XX/YY$ measurement channels that the 2D grid circuit does not have (it measures only $Z$, $X$, $ZZ$). Both surrogates share the same local/positional input convention, so the comparison is apples-to-apples. We run the full held-out reconstruction and downstream-classification battery (5 seeds, stratified splits) across CIFAR-10, Fashion-MNIST, and PathMNIST.

The broader surrogate sweep reproduces the real-circuit pattern: HVK1D-pair and HVK2D-pair track each other closely on every dataset and metric (PSNR differs by at most 0.07 dB, accuracy by at most 2 percentage points), and both sit slightly below the raw-linear control. For inference, held-out image differences are first averaged within each split seed so that images sharing a fitted readout are not treated as independent replicates. The seed-level HVK1D-minus-HVK2D PSNR differences are $-0.070$ dB on CIFAR-10 (bootstrap 95% CI $[-0.151, 0.024]$, Wilcoxon $p = 0.3125$), $-0.021$ dB on Fashion-MNIST (CI $[-0.033, -0.006]$, $p = 0.125$), and 0.003 dB on PathMNIST (CI $[-0.012, 0.015]$, $p = 0.625$), all at $n = 5$ seeds. The Fashion-MNIST percentile interval excludes zero, but the discrete seed-level signed-rank test does not resolve the difference; we therefore describe it as numerically small rather than statistically established. Neither topology shows a general reconstruction/classification advantage over simple classical baselines, and the evidence supports treating topology as a resource-cost choice at the tested scale. Section 3.8's restricted, leakage-audited diagnostic remains the setting where entanglement structure matters within the tested feature/readout class.

# 5 Held-Out Output-Level $D_4$ Symmetry Evaluation

The main paper's $D_4$-equivariance claim (pooled HVK2D observable map, equivariance error $9.57\times 10^{-17}$) was previously demonstrated at feature-grid level on 1,000 cached CIFAR-10 images, giving 7,000 image–nonidentity-transform evaluations. That large numerical check did not use a fitted readout, a held-out reconstruction protocol, or a classical baseline sharing the same pooling construction, and therefore did not measure what equivariance costs or buys at the output level. This section closes those gaps with a held-out, multi-seed (5 seeds, 20 train / 10 held-out images per seed, disjoint CIFAR-10 splits) experiment across five modes: the exactly-equivariant-by-construction *D4-pooled-HVK2D* map; a new *D4-pooled-classical* baseline applying the identical Reynolds-averaging construction (group-averaging the feature map over the eight $D_4$ transforms) to the purely classical *local-raw* feature map instead of the HVK2D-surrogate one, so the comparison is architecture-matched rather than quantum-vs-nothing; the non-symmetric *HVK2D-positional* control; an *ordinary-augmentation* baseline (*HVK2D-augmented*: the same non-pooled

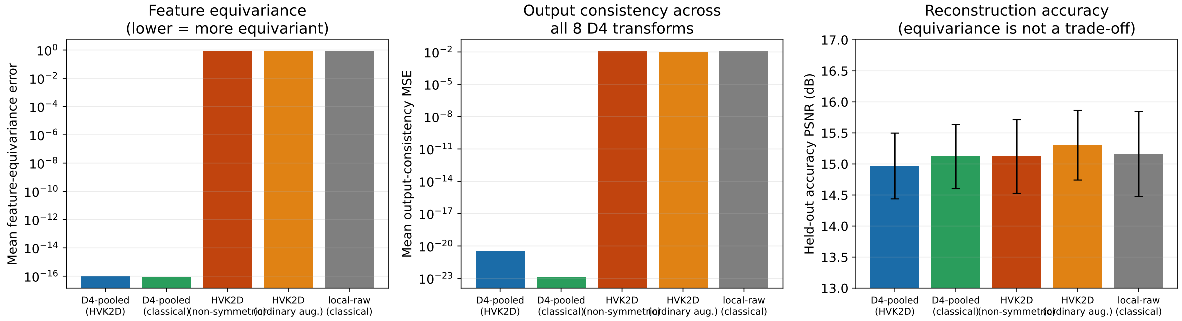


Figure 7: Held-out output-level $D_4$ symmetry evaluation, 5 seeds. Left: feature-equivariance error (log scale) — both pooled modes sit at floating-point precision, roughly 16 orders of magnitude below every non-pooled control. Center: output-consistency MSE across all 8 $D_4$ transforms (log scale) — the same separation persists at the reconstruction-output level, not just the feature level. Right: held-out accuracy PSNR (linear scale, error bars = std) — the five mean values are close relative to their between-seed variation; no formal equivalence claim is made.

feature map, but with a ridge readout trained on all eight $D_4$-transformed copies of every training image, instead of architectural pooling); and the non-symmetric classical *local-raw* control. For every mode we measure both *feature-equivariance error* (as before) and, new here, *output consistency*: for each held-out image we predict a reconstruction from every one of its 8 $D_4$-transformed copies, rotate/reflect each prediction back to canonical orientation, and measure the pairwise mean-squared disagreement between the 8 canonicalized predictions, together with reconstruction accuracy (PSNR) against ground truth.

Table 13: Held-out output-level $D_4$ symmetry evaluation (5 seeds, 10 held-out images/seed). Feature-equivariance error and output-consistency MSE are reported at ±std over seeds; both span many orders of magnitude between pooled and non-pooled modes.

| Mode | Equivariance error | Output consistency (MSE) | Accuracy PSNR (dB) |
|---|---|---|---|
| D4-pooled (HVK2D) | $(9.47 \pm 0.25) \times 10^{-17}$ | $(3.19 \pm 1.66) \times 10^{-21}$ | $14.97 \pm 0.53$ |
| D4-pooled (classical) | $(8.94 \pm 0.22) \times 10^{-17}$ | $(1.37 \pm 0.96) \times 10^{-23}$ | $15.12 \pm 0.52$ |
| HVK2D (non-symmetric) | $0.8163 \pm 0.0070$ | $0.01149 \pm 0.00213$ | $15.12 \pm 0.59$ |
| HVK2D (ordinary aug.) | $0.8163 \pm 0.0070$ | $0.01055 \pm 0.00226$ | $15.30 \pm 0.56$ |
| local-raw (classical) | $0.8380 \pm 0.0067$ | $0.01202 \pm 0.00245$ | $15.16 \pm 0.68$ |

Four findings follow. First, the Reynolds-averaging construction that makes HVK2D exactly equivariant by construction is not specific to the quantum-surrogate feature map: applying the identical pooling operation to a purely classical local-observable map produces the same floating-point-precision equivariance ($8.94 \times 10^{-17}$ vs. $9.47 \times 10^{-17}$), confirming the guarantee is a property of the group-averaging construction itself, not of anything intrinsically quantum. Second, the separation between pooled and non-pooled modes is not an artifact of measuring features in isolation: it persists essentially unchanged at reconstruction output (output-consistency MSE $\sim 10^{-21}$–$10^{-23}$ for pooled modes vs. $\sim 0.010$–$0.012$ for non-pooled controls). Third, ordinary augmentation cannot change feature-equivariance error, but it reduces output-consistency MSE by roughly 8% (0.01055 vs. 0.01149). Fourth, exact pooling has a small accuracy trade-off relative to ordinary augmentation: pooled HVK2D is lower by 0.336 dB on average across the five paired seeds (seed-bootstrap 95% CI $[-0.441, -0.231]$ dB; two-sided Wilcoxon $p = 0.0625$, the smallest attainable nonzero two-sided value at $n = 5$). The symmetry property is exact by construction — a design-correctness check on the pooling operator, not a quantum capability — while the available sample supports a modest rather than zero accuracy cost.

### 5.1 Real-Circuit Confirmation

Everything above is measured on the closed-form analytic surrogate. This subsection repeats the pooled-vs-non-pooled equivariance and output-consistency comparison on the *actual trained quantum circuit* — the same four already-trained HVK2D checkpoints used for the main paper's real IBM-hardware pilot (*cat*, *ship/hydrofoil*, *ship/sea boat*, *airplane*), replayed with no retraining. For each image we run the trained model's real PennyLane forward pass (exact statevector simulation of the trained weights, not the surrogate formula) on all eight $D_4$-transformed copies of the full image, and compute the same feature-equivariance error and output-consistency MSE as above, both without pooling and with the identical Reynolds-averaging construction applied to the real circuit's own output grid.

Across all four checkpoints the result is unambiguous. Without pooling, the mean feature-equivariance error is 0.2777 and the mean output-consistency MSE is $4.355 \times 10^{-2}$; with the identical Reynolds-averaging construction applied to the real circuit's own output grid, these fall to $5.39 \times 10^{-8}$ and $1.46 \times 10^{-15}$ respectively — the latter at the numerical floor of double precision. Per-image values for all four checkpoints are in the machine-readable archive. The equivariance is therefore exact by construction on the trained circuit, not only on the analytic surrogate.

The real-circuit result confirms the surrogate's finding directly: pooling collapses the equivariance error from $\mathcal{O}(10^{-1})$ to $\mathcal{O}(10^{-8})$ and output-consistency MSE from $\mathcal{O}(10^{-2})$ to $\mathcal{O}(10^{-15})$, an 8–13 order-of-magnitude gap on every one of the four images, with no exceptions. The pooled real circuit's residual error ($\sim 5 \times 10^{-8}$) is larger than the surrogate's floating-point-precision result ($\sim 10^{-17}$) because the real circuit involves a full statevector simulation of the trained gates plus a fresh per-transform feature-standardization step, both of which accumulate ordinary floating-point noise that the closed-form surrogate formula does not — the pooled result is exact up to numerical precision, not exact to the same sixteen digits as a symbolic identity, which is the expected and unremarkable difference between an analytic proxy and an actual simulated circuit.

## 6 Exact Circuit and Hamiltonian Definitions

For a one-dimensional chain of $n_{\text{bonds}}$ bonds indexed by $i$ and sites $(i, i+1)$, the anisotropic Heisenberg Hamiltonian is

$$H_{\text{Heis}} = \sum_{i=0}^{n_{\text{bonds}}-1} \Big( J_x^{(i)} X_i X_{i+1} + J_y^{(i)} Y_i Y_{i+1} \\ + J_z^{(i)} Z_i Z_{i+1} \Big). \tag{4}$$

Here $X, Y, Z$ are Pauli operators and $J_\alpha^{(i)}$ are bond-resolved couplings. HVK1D fixes six sites and treats the 15 coefficients on its five bonds as learnable parameters initialized independently from a zero-mean Gaussian with standard deviation 0.1. Given patch $n$'s measured observables, its energy diagnostic is

$$\mathcal{E}_n(\theta, J) = \sum_{i=0}^{4} \sum_{\alpha \in \{x,y,z\}} J_\alpha^{(i)} \langle \sigma_\alpha^{(i)} \sigma_\alpha^{(i+1)} \rangle_n. \tag{5}$$

HVK2D instead uses the seven edges $\mathcal{G}_{2D}$ of its $2 \times 3$ circuit graph and the implemented grid-Ising diagnostic $\mathcal{E}_n^{2D} = \sum_{(u,v)\in\mathcal{G}_{2D}} j_{uv} \langle Z_u Z_v \rangle_n$. Thus "Hamiltonian diagnostic" refers to the full $XX/YY/ZZ$ chain energy for HVK1D and the learned $ZZ$ graph energy for HVK2D. In either case, the patch-averaged term is $\mathcal{L}_{\text{energy}} = \frac{1}{N}\sum_n \mathcal{E}_n$, and the total objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rec}} + \lambda\,\mathcal{L}_{\text{energy}}, \tag{6}$$

$$\mathcal{L}_{\text{rec}} = \frac{1}{N}\sum_{n=1}^{N} \big\| D(o_n, e_n) - P_n \big\|_F^2. \tag{7}$$

with $D$ the classical decoder and $P_n$ the ground-truth patch.

Per-patch coordinates $(r_i, c_j)$ are mapped to Fourier features $e_{4k} = \sin(\omega_k \pi r_i)$, $e_{4k+1} = \cos(\omega_k \pi r_i)$, $e_{4k+2} = \sin(\omega_k \pi c_j)$, and $e_{4k+3} = \cos(\omega_k \pi c_j)$ [9]. HVK1D uses $k \in \{0, 1\}$ (8-D), whereas the native-resolution HVK2D implementation uses one frequency (4-D); a learned linear map projects either vector to six $R_Y$ angles. Both six-qubit circuits first apply $U_{\text{enc}}(x_n) = \bigotimes_{i=0}^{5} R_X(x_n^{(i)})$ and $U_{\text{pos}} = \bigotimes_{i=0}^{5} R_Y(\theta_{\text{position}}^{(i)})$. Each of two HVK1D layers then applies per-qubit Rot gates followed by the strongly-entangling CNOT ring. Each HVK2D layer applies CNOTs over the four horizontal and three vertical grid edges followed by per-qubit Rot gates, matching the executed source and hardware-replay circuits.

HVK1D returns local $Z/X$ measurements and the three nearest-neighbor pair sectors $ZZ/XX/YY$ (27 values); HVK2D returns local $Z/X$ measurements and grid-edge $ZZ$ correlations (19 values). Positional rotations and fixed graph connectivity distinguish circuit sites, so neither unpooled map is asserted to be permutation- or $D_4$-equivariant. The $D_4$ action in the main paper is instead the explicit permutation action on the $4 \times 4$ patch grid, and equivariance follows only after the stated Reynolds average over its eight transformations.

# 7 Hardware Pilot Methodology

The main paper reports a real IBM Quantum hardware reconstruction pilot (five images: *Monalisa* and four CIFAR-10 images, each optimized per-image, hardware PSNR 25.90–31.52 dB against 36.56–46.79 dB in noiseless simulation). This section gives the complete methodology.

## 7.1 Reused Checkpoints

No model was retrained for this pilot. Two pre-existing artifacts were reused directly. The first was the "shared VQC baseline" Monalisa checkpoint under

```
main2/newHVK/results/ablation_study/legacy_hvk_controls/eval_controls/
shared-baseline-seed-42/.
```

This is the run underlying the "Legacy signed energy" row of Table 9. The second set comprised four fresh HVK2D checkpoints trained for 200 steps with Adam ($\eta = 0.004$) on an NVIDIA GTX 1050. Each checkpoint corresponds to one native-resolution $32 \times 32$ CIFAR-10 image optimized directly on its own target, following the same per-image protocol as *Monalisa* rather than the held-out suite of Section 3.3. Non-overlapping $8 \times 8$ patches give 16 patches per image, matching the Monalisa pilot's patch count and keeping the hardware circuit budget predictable.

### 7.2 Circuit Reconstruction

The trained HVK1D forward pass applies, per patch: $R_X$ angle embedding, $R_Y$ positional modulation, then two `StronglyEntanglingLayers` (per-wire $R_Z$-$R_Y$-$R_Z$ rotations with a CNOT ring whose range increments by layer). We rebuilt this gate-for-gate in Qiskit. Because the trained circuit measures $Z$, $X$, $ZZ$, $XX$, and $YY$ (27 observables), and pairwise correlators are recoverable as joint parities of the *same* all-$Z$/all-$X$/all-$Y$ basis measurements that already give the single-qubit marginals, only three measurement circuits per patch are needed. For 16 Monalisa patches this is $16 \times 3 = 48$ circuits. The HVK2D grid circuit is simpler (two layers of fixed grid-edge CNOTs then per-qubit rotations) and measures only $Z$, $X$, $ZZ$ (19 observables, no $XX/YY$), so only two measurement circuits per patch are needed: for four CIFAR images at 16 patches each, $4 \times 16 \times 2 = 128$ circuits.

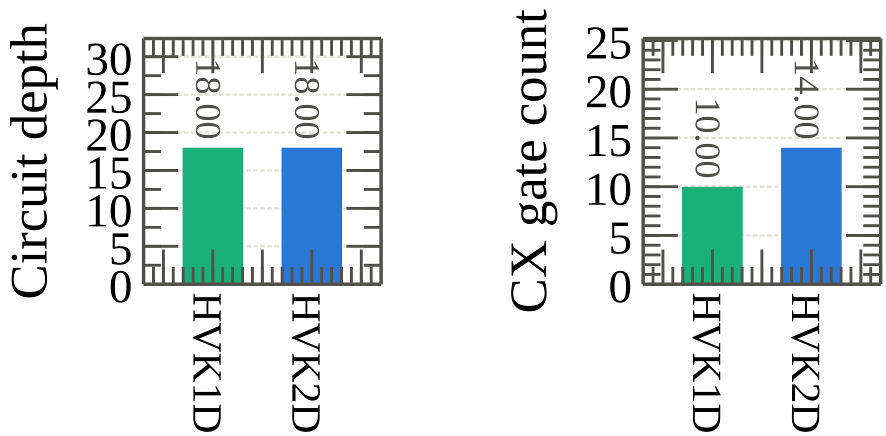


Figure 8: Transpiled circuit resource comparison between the HVK1D (*Monalisa*) and HVK2D (CIFAR) per-patch circuits at `optimization_level=1` on `ibm_fez`: circuit depth (left) and CX-gate count (right). Both circuits have depth 18 and use 10–14 CX gates. These resource counts document the executed circuits; they do not by themselves attribute the observed hardware–simulator PSNR gap to a particular noise source.

### 7.3 Local Validation Before Any Hardware Submission

Every circuit was validated against the cached, training-time observable vectors via an exact Qiskit statevector simulation, *before* any job was submitted to real hardware. Sixteen patches were validated per checkpoint across all five checkpoints; the maximum absolute reconstruction error was $7.89 \times 10^{-4}$ for Monalisa (HVK1D) and $7.14 \times 10^{-6}$–$9.42 \times 10^{-6}$ for the four CIFAR HVK2D checkpoints.

All translation discrepancies are small relative to the observable perturbations induced by finite-shot device execution: across the five checkpoints, the hardware–statevector mean absolute observable deviation is 0.096–0.125 (maximum 0.218–0.340), versus a maximum translation discrepancy of $7.89 \times 10^{-4}$ for Monalisa and $9.42 \times 10^{-6}$ for CIFAR. They therefore rule out a gate-translation mismatch of the measured magnitude as the primary explanation for the hardware–simulator PSNR gap. This pilot does not separate sampling noise, gate/readout errors, drift, and transpilation effects.

### 7.4 IBM Quantum Account, Backend, and Job Identifiers

Jobs were submitted via `qiskit-ibm-runtime` (channel `ibm_quantum_platform`) under a free/open-plan instance. Both pilots ran on `ibm_fez`, a 156-qubit Heron-class processor. Transpilation used `optimization_level=1`; each job used `SamplerV2` with 256 shots per circuit.

Table 14: IBM Quantum job identifiers (backend: `ibm_fez`, 256 shots/circuit).

| Image | Job ID | Circuits | Bases |
|---|---|---|---|
| Monalisa | `d9ecu34inv1c73aq3qt0` | 48 | Z, X, Y |
| CIFAR cat | `d9edfo2neu4c739ob2ig` | 32 | Z, X |
| CIFAR ship (hydrofoil) | `d9edg5cjeosc73filhng` | 32 | Z, X |
| CIFAR ship (sea boat) | `d9edgakjeosc73filhug` | 32 | Z, X |
| CIFAR airplane | `d9edgdphtsac739e4kdg` | 32 | Z, X |

Total: 176 circuits across 5 jobs. Quota consumption, read from `QiskitRuntimeService.usage()` immediately before and after the pilot, changed from 19 to 35 seconds against a 600-second monthly free-plan allowance. Thus the account meter indicated 16 seconds of billed quantum-processor time and 565 seconds remaining afterward, at an observed rate of $\approx 0.09$–$0.13$ s/circuit. The readings were recorded contemporaneously, but the raw service-response object was not serialized; unlike the job identifiers and reconstruction outputs, this quota figure is therefore an execution-log record rather than a repository-recomputable artifact.

### 7.5 Repeated-Execution Anchor Jobs

The main paper's repeated-execution anchor points — the same trained checkpoints, replayed without retraining on two backends other than `ibm_fez` — were four further jobs, listed in Table 15 so that every real-hardware number in the main text has a job identifier here. Submission settings match the pilot above (`SamplerV2`, `optimization_level=1`); only backend and shot count differ. The accompanying shot-budget sweep reported alongside them in the main text is a calibrated device-noise *simulation* (`qiskit-aer` with the `FakeFez` calibration snapshot) and consumed no quantum-processor time, so it has no job identifier.

Table 15: Anchor-job identifiers for the repeated-execution checks. Job identifiers, backends, shot counts and PSNR values are read from the retained execution record `IBM_Cloud/outputs/hardware_robustness_study/real_hardware_anchors.json`; the PSNR column matches the main text.

| Image | Backend | Job ID | Shots | Circuits | PSNR (dB) |
|---|---|---|---|---|---|
| Monalisa (HVK1D) | `ibm_marrakesh` | `d9gqm58gk0ls73f219m0` | 256 | 48 | 25.94 |
| Monalisa (HVK1D) | `ibm_kingston` | `d9gqnk0gk0ls73f21bkg` | 1024 | 48 | 26.10 |
| CIFAR cat (HVK2D) | `ibm_kingston` | `d9gqqi0gk0ls73f21glg` | 256 | 32 | 28.93 |
| CIFAR cat (HVK2D) | `ibm_kingston` | `d9gqr8chonhs73ac3fmg` | 1024 | 32 | 31.24 |

### 7.6 Cross-Platform and Bond-Dimension Replay Job Ledger

Three further checkpoint-replay campaigns were executed beyond the reconstruction pilot, and are recorded here for auditability only: an IonQ ideal-simulator cross-platform check of the order parameter (48 circuits per job — two images, three widths $N \in \{4, 6, 8\}$, eight checkpoint epochs); a bond-dimension order-parameter replay ($N = 6$, $\chi \in \{1, 2, 4, 8\}$, 64 circuits per job) on real `ibm_marrakesh` hardware and on the IonQ ideal simulator; and a three-basis energy-to-entanglement replay over the same checkpoint grid (192 circuits per job). Every job completed and produced non-degenerate curves whose shape matches their noiseless-simulator counterparts. None of them supports a claim in this paper — they are single-patch, single-seed probes of a readout whose threshold fails its own negative control (Section 3.9) — so the curves are archived rather than reproduced here, and Table 16 records the job identifiers instead. The IonQ runs use an ideal cloud simulator, not an IonQ QPU: they test circuit translation, submission, count retrieval, bit-order handling, and estimator reproducibility, and say nothing about device noise or comparative hardware quality.

One construction limit is worth recording. At $\chi = 1$ the MPS is a product state and the archived entropy sum sits at the numerical floor, $S = 5.0 \times 10^{-6}$, against $S = 0.014$–$0.045$ for $\chi \geq 2$; consequently $R_{ES} = H/S$ reaches $O(10^5$–$10^6)$ purely by division by a near-zero denominator. Those rows are retained in the machine-readable archive but excluded from the archived curves — a construction-limit audit, not a physical divergence. Complete per-shot IBM and IonQ summaries are stored under `IBM_Cloud/outputs/checkpoint_bond_dimension_hardware_sweep/` and `IBM_Cloud/outputs/checkpoint_bond_dimension_temperature_hardware_sweep/`.

Table 16: Replay job ledger for the three checkpoint-replay campaigns whose curves are archived rather than reproduced in this document. Job identifiers are given so the executions can be audited; no claim in this paper rests on them.

| Campaign | Platform | Backend | Shots | Job ID |
|---|---|---|---|---|
| Order parameter, cross-platform | IonQ ideal sim. | `ionq_simulator` | 256 | `019f903b-62c0-74db-adec-a33847ae8a1c` |
| Order parameter, cross-platform | IonQ ideal sim. | `ionq_simulator` | 512 | `019f903d-09d4-7548-87f1-2a55185ee9b5` |
| Order parameter, cross-platform | IonQ ideal sim. | `ionq_simulator` | 1024 | `019f903e-cac2-708f-911f-67b2b6d7471c` |
| $R_{ES}$, three-basis | IonQ ideal sim. | `ionq_simulator` | 1024 | `019f9046-7ba6-7320-9766-552d88d490bf` |
| Bond-dim. order parameter | IBM hardware | `ibm_marrakesh` | 256 | `d9hqu0jsbqfc73eqgr80` |
| Bond-dim. order parameter | IBM hardware | `ibm_marrakesh` | 512 | `d9hr3d8gk0ls73f3elsg` |
| Bond-dim. order parameter | IBM hardware | `ibm_marrakesh` | 1024 | `d9hr43l0k0jc738il8ug` |
| Bond-dim. order parameter | IonQ ideal sim. | `ionq_simulator` | 256 | `019f9571-21d0-7339-9111-d31a308440a1` |
| Bond-dim. order parameter | IonQ ideal sim. | `ionq_simulator` | 512 | `019f9573-1ed9-768c-9768-465fe75a409b` |
| Bond-dim. order parameter | IonQ ideal sim. | `ionq_simulator` | 1024 | `019f9574-a981-71dd-94b0-9d8fb8968f49` |
| Bond-dim. $R_{ES}$ | IBM hardware | `ibm_marrakesh` | 256 | `d9hr44shonhs73adgq40` |
| Bond-dim. $R_{ES}$ | IBM hardware | `ibm_marrakesh` | 512 | `d9hr5u50k0jc738ilbs0` |
| Bond-dim. $R_{ES}$ | IBM hardware | `ibm_marrakesh` | 1024 | `d9hrb8shonhs73adh8mg` |
| Bond-dim. $R_{ES}$ | IonQ ideal sim. | `ionq_simulator` | 256 | `019f957f-dfbe-70fc-b6aa-b2398317f049` |
| Bond-dim. $R_{ES}$ | IonQ ideal sim. | `ionq_simulator` | 512 | `019f9584-4ba3-7224-ab7b-e81e32e6c1e2` |
| Bond-dim. $R_{ES}$ | IonQ ideal sim. | `ionq_simulator` | 1024 | `019f9586-4eb6-7596-ba47-dc6132c7f943` |

### 7.7 Reconstruction Decode

Hardware counts were converted to observable expectation values by the standard parity estimator, $\langle Z_i \rangle = \frac{1}{S}\sum_{\text{shots}}(-1)^{b_i}$ (identically for $X$, $Y$ after basis rotation), with pairwise correlators computed as the shot-weighted mean of $(-1)^{b_i+b_j}$ within the same measurement basis. The resulting observable vector, concatenated with the training-time positional encoding, was passed through the *unmodified* trained decoder. No hardware-specific fine-tuning, error mitigation, or post-hoc calibration was applied.

## 8 Reproducibility and Artifact Map

The repository-level experiment drivers set Python, NumPy, and PyTorch random seeds explicitly. Unless a section states otherwise, multi-seed studies use seeds $\{0, 1, 2, 3, 4\}$; the reduced real-circuit topology confirmation and six-dataset reconstruction extension use $\{0, 1, 2\}$. Held-out samples are disjoint from their corresponding fitting samples, and each reported paired test preserves the pairing unit stated in its table or surrounding text (image within split, or seed after within-seed image aggregation). "Mean ± std" denotes the arithmetic mean and population-standard-deviation summary (NumPy `ddof=0`) over the unit named in the caption. Bootstrap intervals are percentile 95% intervals; Wilcoxon values are two-sided signed-rank tests. No multiple-comparison correction is applied, so $p$-values should be interpreted as individual-comparison summaries rather than as a family-wise discovery analysis.

The current verification environment uses Python 3.11.9, PyTorch 2.6.0 with CUDA 12.4, PennyLane 0.42.3, Qiskit 2.5.0, NumPy 2.4.6, pandas 3.0.3, scikit-learn 1.9.0, quimb 1.11.2, cotengra 0.7.5, and autoray 0.6.12 on an NVIDIA GeForce GTX 1050. The package constraints are declared in `pyproject.toml`; because that file does not pin every dependency to an exact version, the versions above identify the environment used for this verification pass rather than claiming bitwise reproducibility across arbitrary future installations.

The principal artifact map is as follows. The lowercase driver `main2/newHVK/run_newhvk_suite.py` generates the synthetic restricted-pair diagnostic under `main2/newHVK/results/full_ablation_suite/` and the held-out CIFAR-10 controls under `main2/newHVK/results/q1_validation/`; the six-dataset extension is generated by `main2/newHVK/run_multi_dataset_validation.py` and stored under `main2/newHVK/results/multi_dataset_validation/`. The newer experiment drivers use the repository's historically uppercase path: `Main2/newHVK/run_topology_comparison.py`, `Main2/newHVK/run_d4_symmetry_experiment.py`, and `Main2/newHVK/run_phase_transition_multi_dataset.py`, with outputs respectively under `Main2/newHVK/results/topology_comparison/`, `Main2/newHVK/results/d4_symmetry_experiment/`, and `Main2/newHVK/results/phase_transition_multi_dataset/`. The topology directory includes raw per-run JSON, summaries, paired statistics, and a surrogate manifest. Excluded legacy training-mode traces and any future deterministic evaluation-mode traces have distinct filenames and provenance fields. Path case is intentional and must be preserved on case-sensitive systems. Hardware-pilot result metadata and measured observable arrays are stored under `IBM_Cloud/outputs/hardware_reconstruction/` and `IBM_Cloud/outputs/hvk2d_cifar_hardware_reconstruction/`; the repeated-execution anchor jobs and the calibrated-noise shot sweep under `IBM_Cloud/outputs/hardware_robustness_study/`; bond-dimension hardware replays use the two output directories listed in Section 7.6. Raw and summary artifacts are retained separately so that manuscript means and uncertainty intervals can be recomputed without reading values from figures.

# 9 Discussion

## 9.1 Task-Dependent Representational Scope

The tie-with-classical outcome in Section 3.3 is consistent with a broader pattern in quantum machine learning: whether a quantum feature map or kernel offers an advantage over a classical model depends on the data, target, and comparison class rather than on the model label alone. At the resolution and patch size tested here, the empirical result is that local and raw-linear maps already capture the reconstruction signal available to the shared readout; we do not infer an entanglement property of the image distribution from that observation. The restricted diagnostic of Section 3.8 is the contrasting case: its target is constructed around a distant-element product that the tested raw or local features cannot represent under the fixed linear readout, and the entangling channel separates in that setting. Together, the results delimit the tested operating regime: entangling observables are necessary within the restricted feature/readout class, but are not isolated as necessary for ordinary held-out reconstruction under the evaluated protocol.

### 9.2 Design and Evaluation Guidelines

We distill this ablation study into concrete guidance for evaluating hybrid quantum–classical vision models: (1) *resource-match* every classical control to the quantum model's feature width and readout parameter count; (2) *freeze-isolate* each trainable component independently before crediting any one of them with an observed gain; (3) *audit every favorable diagnostic for target–feature leakage* by tracing target- and feature-construction code for shared subexpressions; (4) *report held-out, not same-set, performance* as the primary evidence for a representation-learning claim; (5) *report multiple seeds with error bars* on any comparison intended to support a quantitative claim.

## 10 Conclusion

The resource-matched, multi-seed study delineates the tested operating regime of HVK. Under a shared fitted readout, simple local/raw maps match or exceed the HVK2D pair-observable map on ordinary held-out reconstruction. The entangling observable channel separates on the restricted target constructed to require nonlocal correlations; separately, $D_4$ pooling enforces equivariance to numerical precision but does not improve held-out reconstruction accuracy in the tested experiment. On held-out CIFAR-10, local-observable and raw-linear controls reach $18.80 \pm 1.42$ dB compared with HVK2D's $18.12 \pm 1.54$ dB; seed-level inference gives a $-0.68$ dB HVK2D-minus-control difference (bootstrap 95% CI $[-0.91, -0.46]$ dB; Wilcoxon $p = 0.0625$, $n = 5$). The same numerical direction appears on five further real datasets. These results motivate task-aligned HVK variants and strengthen the framework's evidentiary scope. The leakage and provenance audits contribute a transferable validation protocol, and Section 7 provides the complete hardware-pilot methodology.

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